SANJEEVI M

AI/ML Data Engineer

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PROFESSIONAL SUMMARY

AI/ML Data Engineer with 4.5+ years of enterprise experience specializing in cloud data platform automation (Snowflake, Databricks, Delta Lake, PySpark) and applied GenAI/agentic tooling (Anthropic Claude API, multi-agent systems, tool-use/function calling). Proven expertise architecting metadata-driven dynamic data masking frameworks auto-generating 124+ role-aware Snowflake views and deploying resilient multi-agent platforms with circuit breakers and backoff retries. Comfortable owning high-impact initiatives from analysis through POC, production deployment, and executive stakeholder demos.

TECHNICAL EXPERTISE

- **AI / GenAI & Agentic Systems:** LLMs, Anthropic Claude API, Multi-Agent Orchestration (Supervisor-Specialist), Tool-Use / Function Calling, LangChain, LlamaIndex, RAG (Retrieval-Augmented Generation), Databricks Vector Search, Embeddings, Prompt Engineering, Agent Memory, ML / NLP, MLOps (MLflow).
- **Data Platforms & Lakehouse:** Snowflake (Stored procedures, dynamic SQL, Masking policies, Row access policies), Databricks (Jobs, Delta Lake, notebooks, Unity Catalog), ADLS Gen2, Apache Iceberg, Kafka.
- **Languages:** Python, SQL, PySpark.
- **Integration & DevOps:** REST APIs (Azure DevOps, SharePoint / Microsoft Graph, GitHub), HashiCorp Vault, ADF (Azure Data Factory), Docker, Git, CI/CD.
- **Practices & Governance:** Data Governance, Data Quality Management, Metadata Management, Data Security & Access Controls (PII Masking), SDLC, Agile / Scrum.
- **BI & Visualization:** Tableau Desktop / Server, Streamlit.

VERIFIED CERTIFICATIONS

- **Claude Certified Architect Foundations** — Verify on Credly
- **Databricks Certified Generative AI Engineer Associate** — Verify on Databricks
- **Databricks Certified Machine Learning Engineer Associate** — Verify on Databricks
- **Databricks Certified Data Engineer Professional** — Verify on Databricks

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions (Client: Humana) | *Oct 2025 – Present*

AI/ML Data Engineer — ODMP (Operational Data Management Platform)

Snowflake Provisioning & Dynamic Field-Masking Automation

- **Metadata-Driven Masking Framework:** Designed and deployed a dynamic masking engine via a Python stored procedure (SP_GENERATE_MASKED_VIEW) reading role/entitlement config rules to auto-generate secure, role-aware Snowflake views with column masking and row-level filtering.
- **Enterprise Scale (POC to 124+ Views):** Scaled framework from a single-view POC to 124 production views across the Sales Consumer schema, handling complex wide schemas with 160+ columns.
- **Automated Provisioning Ingestion:** Built the ACCOUNT_PROVISIONING and FIELD_MASKING data models with a Databricks ingestion pipeline from business SharePoint Excel files (via Microsoft Graph/REST API) into Snowflake with automated IT email alerts.
- **Data Quality & Governance:** Triaged and resolved production schema drift, DDL mismatches, and role-masking validation defects; conducted milestone architecture demos for product owners and analytics leads.

Data Acquisition & Cross-Platform Lakehouse Integration

- **End-to-End API Acquisition Pipeline:** Delivered a compliant API pipeline acquiring 10-year historical and incremental Call Simulator training data into Snowflake (staging → target, compliance schema, DGO-certified source) and authored the Interface Control Document (ICD).
- **Delta Lake Hydration (Microsoft Fabric POC):** Hydrated priority fact tables from Snowflake into Databricks Delta tables, including a dedicated round powering a Microsoft Fabric AI Analytics POC.
- **MADL Access Governance:** Led analysis and access provisioning for Member Admin Data Lake (MADL) policy data, orchestrating intake requests, cross-team approvals, and Databricks table readiness for downstream analytics.
- **Architecture POCs & Production Triage:** Conducted architectural POCs on Apache Iceberg integration for Snowflake ODMP domains and sourcing Milestone Tracker data from external DEP systems; resolved UAT reporting pipeline defects (e.g., Genesys enriched voice mismatches).

PROFESSIONAL EXPERIENCE (CONT.)

Infosys Ltd (Clients: Nike and Apple) | *May 2022 – Oct 2025*

Technology Analyst

- Designed and maintained end-to-end ETL pipelines in Azure Databricks processing large-scale enterprise data for Nike and Apple.
- Engineered Change Data Capture (CDC) pipelines for customer billing and sales orders achieving <1% latency.
- Built automated reporting pipelines eliminating manual Tableau refreshes across multiple operational units.
- Developed real-time anomaly detection dashboards for Apple's Data Operations team with automated Slack alerts for incident escalation.

KEY PROJECTS & INITIATIVES

- **ODMP Multi-Agent Engineering Assistant** | *Claude API, Multi-Agent, Python, Azure DevOps, Databricks*
  Built a Claude API-powered multi-agent assistant enabling natural-language platform operations across 5 engineering systems (Azure DevOps, Git, Databricks, ADLS, SharePoint). Designed with a Supervisor-Specialist architecture, multi-turn memory, and 7 production-hardened resilience patterns including circuit breakers and exponential backoff.
- **Snowpark Rule Framework** | *Humana Inc.* | *Snowpark, Snowflake, JSON*
  Config-driven Snowpark transformation engine dynamically parsing rules from JSON specifications, dramatically reducing manual SQL boilerplate and maintenance overhead.
- **Extract Framework** | *Humana Inc.* | *PySpark, Databricks, Unity Catalog, ADLS Gen2*
  Parameter-driven PySpark framework to extract Snowflake datasets into Delta format on ADLS Gen2 with zero pipeline code modifications.
- **Real-Time E-Commerce Data Lakehouse** | *Apache Iceberg, Kafka, Spark Streaming, MinIO, Airflow, Great Expectations*
  Engineered a medallion architecture (Bronze/Silver/Gold) streaming lakehouse with automated schema validation and data quality assertions. (GitHub)
- **Real-Time Analytics Pipeline** | *Kafka, PySpark, Elasticsearch, Kibana*
  Implemented high-throughput streaming analytics pipeline ingesting e-commerce clickstream and transaction events with sub-second Elasticsearch indexing. (GitHub)
- **Career Flow Engine** | *Python, Selenium, BeautifulSoup, Pandas*
  Automated job market intelligence scraper monitoring job openings across 15+ global markets with personalized alert filtering. (GitHub)

EDUCATION

Post Graduate in Statistics | Presidency College, Chennai (May 2019)